Tobias Hartl

Contact

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Research Interests

Econometrics, State Space and Factor Models, Long Memory, Macro- and Climate Econometrics

Professional Experience

Maastricht University, School of Business and Economics
- Assistant Professor (tenured) of Econometrics 2024-

University of Regensburg, Economics Dept.
- Postdoctoral Researcher (75%) 2023-2024
- Doctoral Researcher (75%) 2017-2023
- Student Assistant 2016-2017

Institute for Employment Research (IAB) Nuremberg, Macroeconomics Dept.
- Postdoctoral Researcher (25%) 2023-2024
- Doctoral Researcher (25%) 2017-2023
- Research Intern 2016

Princeton University, Economics Dept.
- Visiting Postdoctoral Researcher 2023-2024

Education

Dr. rer. pol. (Ph.D.) in Economics, University of Regensburg 2023
- Title: Fractional unobserved components and factor models: econometric theory and applications
- Supervision: Prof. Rolf Tschernig and Prof. Uwe Hassler
- Final Grade: Summa cum laude

M.Sc. in Economics, University of Regensburg 2017
- Final Grade: 1.03 (on a scale of 1-5; 1=best)

B.Sc. in Economics, University of Regensburg 2015
- Final Grade: 1.81 (on a scale of 1-5; 1=best)

Peer-Reviewed Publications

Hartl, T. (2026). Robust trend estimation for strongly persistent data with unobserved memory. *Journal of Business & Economic Statistics*, 44(1):242–254. https://doi.org/10.1080/07350015.2025.2520858

Hartl, T., Hutter, C., and Weber, E. (2026). Matching for three: big data evidence on search activity of workers, firms, and employment service. *Economic Modelling*, 155:107434. https://doi.org/10.1016/j.econmod.2025.107434

Haimerl, P. and Hartl, T. (2023). Modeling COVID-19 infection rates by regime-switching unobserved components models. *Econometrics*, 11(2):10. https://doi.org/10.3390/econometrics11020010

Hartl, T. and Jucknewitz, R. (2023). Multivariate fractional components analysis. *Journal of Financial Econometrics*, 21(3):880–914. https://doi.org/10.1093/jjfinec/nbab022

Hartl, T. and Jucknewitz, R. (2022). Approximate state space modelling of unobserved fractional components. *Econometric Reviews*, 41(1):75–98. https://www.tandfonline.com/doi/full/10.1080/07474938.2020.1841444

Working Papers

Hartl, T. (2020). Macroeconomic forecasting with fractional factor models. Working paper. https://arxiv.org/abs/2005.04897

Hartl, T., Tschernig, R., and Weber, E. (2020b). Fractional trends in unobserved components models. Working paper. https://arxiv.org/abs/2005.03988

Hartl, T., Tschernig, R., and Weber, E. (2020c). Solving the unobserved components puzzle: A fractional approach to measuring the business cycle. Working paper. http://tobiashartl.github.io/files/Hartl_Tschernig_Weber_Puzzle.pdf

Policy Reports

Wanger, S., Hartl, T., and Hummel, M. (2022). Überarbeitung der IAB-Arbeitszeitrechnung im Rahmen der VGR-Sommerrechnung 2022. *IAB-Forschungsbericht*. https://dx.doi.org/10.48720/IAB.FB.2213. (Title translation: Revision of the IAB working time measurment concept 2022)

Bauer, A., Hartl, T., Hutter, C., and Weber, E. (2021). Search processes on the labor market during the covid-19 pandemic. *CESifo forum*, 22(4):15–19. https://ideas.repec.org/a/ces/ifofor/v22y2021i04p15-19.html

Donsimoni, J. R., Glawion, R., Hartl, T., Plachter, B., Timmer, J., Wälde, K., Weber, E., and Weiser, C. (2020). Covid-19 in Deutschland - Erklärung, Prognose und Einfluss gesundheitspolitischer Maßnahmen. *Perspektiven der Wirtschaftspolitik*, 21(3):250–262. https://dx.doi.org/10.1515/pwp-2020-0019. (Title translation: Covid-19 in Germany: Explanation, Forecasts, and Impact of public health measures)

Hartl, T., Hutter, C., and Weber, E. (2020a). Neueinstellungen in der Krise. *Makronom.de*, 2020-06-18. https://makronom.de/corona-arbeitsmarkt-auswirkungen-neueinstellungen-in-der-krise-36315. (Title translation: New hires during the crisis)

Anger, S., Bauer, A., Bossler, M., ..., Hartl, T., ..., and Zika, G. (2020). Befunde der IAB-Forschung zur Corona-Krise - Zwischenbilanz und Ausblick. *IAB-Forschungsbericht*. https://doku.iab.de/grauepap/2020/IAB_Befunde_COVID-19.pdf. (Title translation: IAB research results on the Corona-crisis)

Hartl, T., Wälde, K., and Weber, E. (2020d). Measuring the impact of the German public shutdown on the spread of COVID-19. *VoxEU.org*, 2020-04-14. https://voxeu.org/article/measuring-impact-german-public-shutdown-spread-covid-19

Hartl, T., Wälde, K., and Weber, E. (2020e). Measuring the impact of the German public shutdown on the spread of COVID-19. *Covid Economics*, 1:25–32. https://cepr.org/publications/covid-economics-issue-1

Hartl, T. and Weber, E. (2020). Welche Maßnahmen brachten Corona unter Kontrolle? *Ökonomenstimme*, 2020-05-12. (Title translation: Which measures brought Corona under control?)

Wanger, S., Hartl, T., and Zimmert, F. (2019). Revision der IAB-Arbeitszeitrechnung 2019. *IAB-Forschungsbericht*. https://doku.iab.de/forschungsbericht/2019/fb0719.pdf. (Title translation: Revision of the IAB working time measurment concept 2019)

Third-party Funded Project

NWO Veni Grant, Dutch Research Council 2026-2029
Project "UnveilML: Unveiling latent time series using machine learning: " (309,000 EUR, ranked # 1 out of 25 within cluster)

DFG Research Grant, German Research Foundation 2021-2024
Co-authored the proposal for project "Multivariate fractional unobserved components and factor models for macroeconomic analysis and forecasting" (234,000 EUR; Applicants: Prof. Rolf Tschernig and Prof. Enzo Weber)

Awards and Scholarships

Alumnus of the Lindau Nobel Laureate Meeting 2020

Award for Best Master's Thesis, Economics Dept., University of Regensburg 2017

Award for Best Master's Degree, Economics Dept., University of Regensburg 2017

Student Scholarship, Friedrich-Ebert-Stiftung 2013-2017

Presentations at Conferences

The Netherlands Econometric Study Group (Tilburg University) 2026

Department Seminar (Vrije Universiteit Amsterdam, invited) 2025
Department Seminar (Passau University, invited)
Department Seminar (Regensburg University, invited)

Department Seminar (Maastricht University, invited) 2024
Workshop on High-dimensional Time Series in Macroeconomics and Finance (IHS Vienna)
Int. Association of Applied Econometrics Annual Conference (Thessaloniki)
Aarhus Workshop on Econometrics (Aarhus, invited)

Int. Association of Applied Econometrics Annual Conference (BI Norwegian Business School) 2023
Int. Conference on Computational and Methodological Statistics (HTW Berlin, invited)

Symposium of the Society for Nonlinear Dynamics and Econometrics (virtual) 2022
Department Seminar (University of Regensburg)
Workshop on High-dimensional Time Series in Macroeconomics and Finance (IHS Vienna)
Annual Meeting of the German Economic Society (University of Basel)
Annual Meeting of the German Statistical Society (University of Münster)
Int. Conference on Computational and Methodological Statistics (King's College London, invited)

Seminar on Long Memory Econometrics (virtual, invited) 2021
Department Seminar (University of Regensburg)
European Summer Meeting of the Econometric Society (virtual)
Annual Meeting of the German Statistical Society (virtual)
Annual Meeting of the German Economic Society (virtual)
Int. Conference on Computational and Methodological Statistics (virtual)

World Congress of the Econometric Society (virtual) 2020
Symposium of the Society for Nonlinear Dynamics and Econometrics (virtual)
Department Seminar (University of Regensburg)
PhD Seminar (University of Regensburg)

Int. Conference on Computational and Methodological Statistics (virtual)

Department Seminar on Statistics and Econometrics (Kiel University, invited) 2019
Econometric Seminar (IAB Nuremberg)
Joint Statistical Meeting of the German Statistical Society (LMU Munich)
Workshop on High-dimensional Time Series in Macroeconomics and Finance (IHS Vienna)
Annual Meeting of the German Statistical Society (Trier University)
Annual Meeting of the German Economic Society (Leipzig University)
Int. Conference on Computational and Methodological Statistics (University of London)

Long Memory Conference (Aalborg University) 2018
Workshop on Statistics and Econometrics (University of Passau)
European Summer Meeting of the Econometric Society (University of Cologne)
Annual Meeting of the German Statistical Society (JKU Linz)
Int. Conference on Computational and Methodological Statistics (University of Pisa)

Referee Service

Referee for Advances in Statistical Analysis, Empirical Economics, International Journal of Forecasting, Journal of Applied Econometrics, Journal of Business & Economic Statistics, Journal of Labour Market Research, Journal of Official Statistics, Journal of Quantitative Economics, Journal of Time Series Analysis, Statistical Papers, Statistics, World Development

Grant application reviewer for German Research Foundation (DFG)

Teaching Experience

Statistics (Lecture and Tutorial), Maastricht University Summer 2026
Undergraduate course (6.5 ECTS). The first-year course for Business Engineering students provides an introduction to statistics, covering both descriptive methods as well as inference at the beginner's level. Average student evaluation (on a scale of 10-1; 10=best): **8.9** (2025) Summer 2025

Mathematical Statistics (Tutorial), Maastricht University Summer 2026
Undergraduate course (6.5 ECTS). The second-year course formally introduces students to the principals of statistical inference. Average student evaluation (on a scale of 10-1; 10=best): **8.9** (2026), **9.2** (2025) Summer 2025

Quantitative Methods 3 (Tutorial), Maastricht University Winter 2025
Undergraduate course (4 ECTS). The skills course covers the application of basic statistical tools such as t-tests, ANOVA, and regression analysis, in several case studies. Average student evaluation (on a scale of 10-1; 10=best): **9.1** (2025), **9.0** (2024) Winter 2024

Empirical Econometrics (Lecture and Tutorial), Maastricht University Winter 2025
Undergraduate course (6.5 ECTS). The course covers methods of applied time series econometrics, such as autoregressive and vector autoregressive models, estimation and inference in such settings. A strong emphasis is put on the implementation in R. Winter 2024
Average student evaluation (on a scale of 10-1; 10=best): **9.5** (2025), **9.4** (2024)

Quantitative Economic Research II (Lecture), University of Regensburg — Summer 2024, Summer 2023, Summer 2022, Summer 2021

Graduate course (6 ECTS). The course covers advanced time series models, their identification, specification, and estimation. Topics covered are simultaneous equation models, (structural) vector autoregression and (structural) vector error correction models.
Average student evaluation (on a scale of 1-6; 1=best): **1.0** (2023); **1.6** (2022); **1.2** (2021)

Programming in R (Lecture), University of Regensburg — Winter 2021

Graduate course (2 ECTS). The course gives an introduction into the programming language R. It covers data analysis and manipulation, flow control, regression, simulation, numerical optimization, bootstrap and efficient programming.
Average student evaluation (on a scale of 1-6; 1=best): **1.3**

Advanced Econometrics (Lecture), University of Regensburg — Summer 2024, Summer 2020

Graduate course (6 ECTS). The course details advanced estimation techniques and analyzes their asymptotic properties. Topics covered are the nonlinear regression model, maximum likelihood estimation, generalized least squares, generalized instrumental variables, and the generalized method of moments.
Average student evaluation (on a scale of 1-6; 1=best): **1.3** (2024); **1.3** (2020)

Advanced Dynamic Econometrics (Lecture), University of Regensburg — Winter 2019

Graduate course (2 ECTS). The interactive course captures state space models and related topics such as filtering, smoothing, and parameter estimation.
Average student evaluation: **not evaluated**

Advanced Econometrics (Tutorial), University of Regensburg — Summer 2021

Graduate course (6 ECTS). See above for the description.
Average student evaluation (on a scale of 1-6; 1=best): **1.0**

Advanced Issues in Econometrics (Tutorial), University of Regensburg — Winter 2020

Undergraduate course (6 ECTS). The course focuses on panel data and limited dependent variables. It covers causality and evaluation studies, pooled cross section analysis, fixed- and random-effects estimators, instrumental variables and two stage least squares, simultaneous equation models, Logit and Probit models, and models for sample selection corrections.
Average student evaluation (on a scale of 1-6; 1=best): **1.1**

Thesis supervision,

Many master's theses supervised: 9; Bachelor's thesis supervised: 5

Language and Computer Skills

Computer Skills: R, Python, Matlab, C, Fortran, Gauss, LaTeX
Languages: German (native), English (fluent), Dutch (basic), French (basic)